

CLOUD COMPUTING

Date _____
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WEEK - 1

Trends In Computing

Distributed

Grid

Clusters

Utility

- * Internet
- * ATM's
- * Workgroups
- * 
 - system 1
 - system 2
 - resources

and computing.
= High performance

Electrical Grid Analogy

We use electricity where we do not require the source or technology application.

Cloud

Located very closely to each other.

Centralized form of computing.

works together as single system

- Low latency
- High reliability

Server

Client

Decentralized

Similarly, we can use resources with having no knowledge of hardware capacities, or owner of hardware, or supplier.

eg: Data centres, weather forecasts

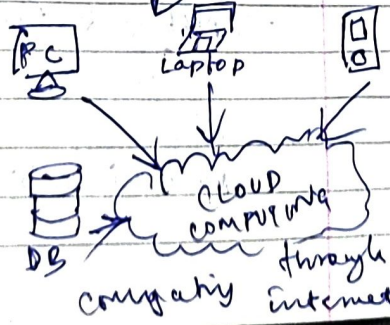
* Decentralized.

Utility

Service provisioning model like Amazon EC2 Billing & services for AWS customers.

* pay for use, pay as you go

Cloud



CRM - software
Customer relationship management

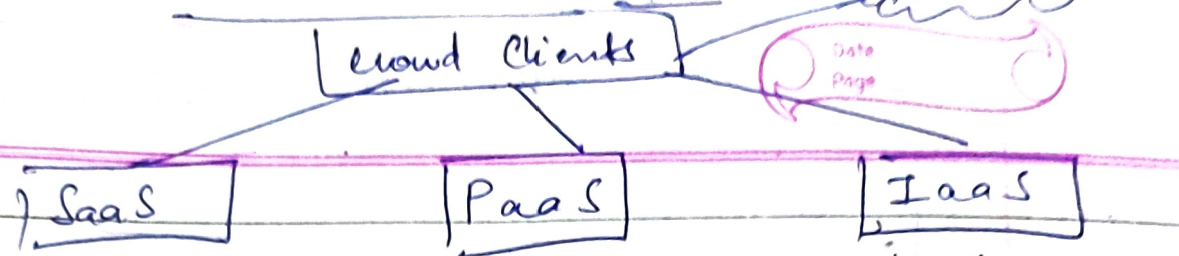
APP
Angry
PAAS

SAAS
Selling, CRM
APP

Client server Model
" " " "
There is no concept of virtualization

Distributed

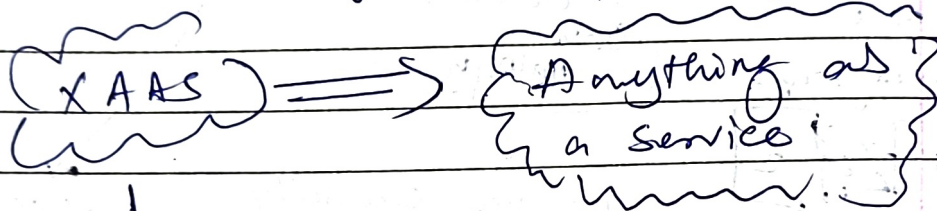
Cloud Service Models :



* Applications are designed for end-user & deployed over web.

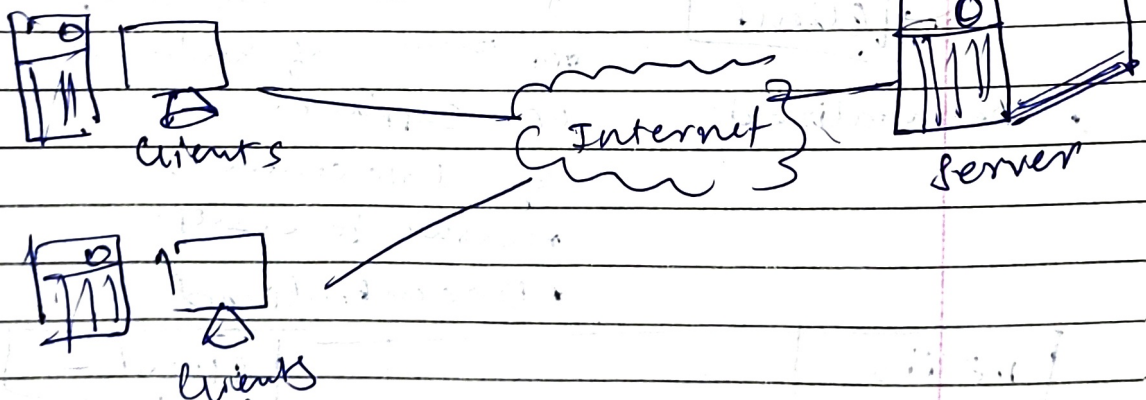
Set of tools & services designed to make coding & deployment of apps faster & efficiently.

Hardware & Software that powers all.



Delivery of any IT function or service over the internet

Client-Server Architecture



Network Function Virtualization

The replacement of network appliance hardware with virtual machines. Eg: Firewalls, Routers.

Deployment models:

Public Cloud

Google App Engine.
MS Windows Azure
Amazon EC2
IBM SMART CLOUD.

Private Cloud

Ubuntu Enterprise Cloud - UEC.
Amazon VPC
VMware, Cloud Infrastructure Suite
MS ECI data centre.

On-site private cloud

Set up on Customer's premises
Eg: office

⊛ Limited Resources

⇒ Has fixed computing & storage capacity.

→ Community Cloud:

⇒ For specific community

Eg: Google Apps for government / any specific organisations like military/army.

⇒ Outsourced: resource provided by an outsider or a vendor.

Hybrid cloud: Mixture of 2 or more clouds

* Such as private, community, public, ...
* Extremely Complex

* ⊛ Lean IaaS Cloud Architecture

* Data Object Storage (DOS)

* Stores subscriber's metadata like credentials, OS images.

* This service is (usually) single for a cloud.

Cluster Manager:

* Responsible of collection of computers that are connected via high speed LAN Cable * Acquires Resource alloc. Commands and queries from the Cloud Manager.

Virtualization:

Ability to run multiple OS on same hardware. Hypervisor = Creates / Manages VM's.

- Learning Approaches to software Virtualization.

* Check OneNote

XML - syntax for "encoding".
Text-based data (words / phrases, numbers).
+ XML parsers
Language rules / have XML processing

→ Defining Language dialects

XML DTD.

XML messaging & processing.

→ Readable by both Humans & machines

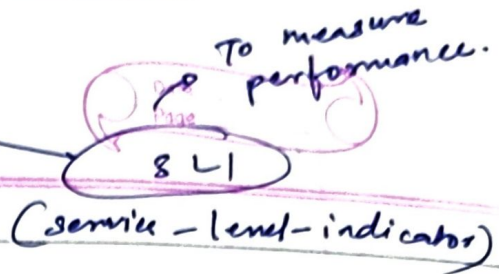
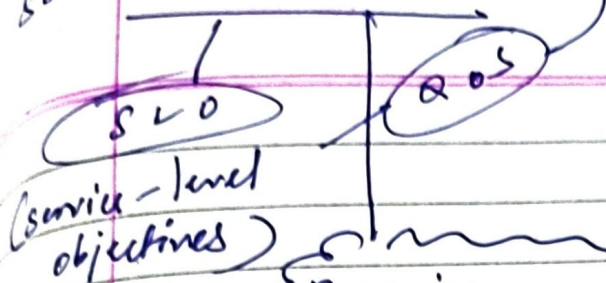
SOAP - Simple Object Access Protocol
Format for sending messages over internet b/w programs. XML-based

UDDI - SOAP-based: Helps users find web services in cloud computing.

WSPL

EDI

Service Level Agreement



Promise
+ Agreement
+ Commitment

$$\Rightarrow \text{Error Budget} = \text{SLA} - \text{SLO}$$

Key-Performance Indicators [KPI]

Availability (A) = $1 - \left(\frac{\text{downtime}}{\text{uptime}} \right)$

coefficient of Variation CV

penalty formula = $0.9 \cdot 8 \cdot \pi \int_0^{\infty} \left| \frac{dD(t)}{dt} \right| dt$

Larger delay in provision.

change in demand

peak computing demand is 120 units

Q. demand function $D(t) = \begin{cases} 50 \sin(t), & 0 \leq t \leq \pi/2 \\ 20 \sin(t), & \pi/2 \leq t \leq \pi \end{cases}$ Given $\delta = \pi/12$

Sol. $D(t) = \int_0^{\pi/2} 50 \sin(t) dt + \int_{\pi/2}^{\pi} 20 \sin(t) dt$

$\frac{d(D(t))}{dt} = 50 \frac{d(\sin(t))}{dt} + 20 \frac{d(\sin(t))}{dt}$

$\frac{dD(t)}{dt} = 50 \int_0^{\pi/2} \cos(t) dt + 20 \int_{\pi/2}^{\pi} \cos(t) dt$

$$\begin{aligned}
 &= 50 \left[\frac{\sin(t)}{\frac{\pi}{2}} \right]_0^{\frac{\pi}{2}} + (-20) \left[-\sin(t) \right]_0^{\frac{\pi}{2}} \\
 &= 50 \left[\sin\left(\frac{\pi}{2}\right) - \sin(0) \right] + (-20) \left[\sin \pi - \sin(0) \right] \\
 &= 50 [1 - 0] + (-20) [0 - 1] = 50 + 20 \\
 &= 70
 \end{aligned}$$

$$\begin{aligned}
 \text{penalty} &= 0.9 \times \delta \times \int_0^{\pi} \frac{dD(t)}{dt} dt \\
 &= 0.9 \times \frac{\pi}{12} \times 70 \\
 &= 63 \times \frac{\pi}{12} \quad \leftarrow \text{Given in question, } \delta = \text{delay.} \\
 &= 63 \times \frac{3.14}{12} = 16.49 \text{ units}
 \end{aligned}$$

Heartbeat communication.

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Relational Database - organises data into tables, columns, rows. With relationships between them.

→ Amazon DynamoDB

→ MapReduce Model

→ Big data analysis model that processes data sets using a parallel algorithm on computer clusters.

Openstack

→ Cloud OS, manages resources such as compute, storage, networking...
→ offers IaaS functionality.

Hypervisor

→ Software that allows multiple OS to run on single physical computer simultaneously.

Availability =

$\frac{\text{uptime}}{\text{uptime} + \text{downtime}}$

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Megha mala

pvt, open stack based cloud

TRICKS Creating VM's, accessing them, and terminating them.

Microsoft Azure: [12-41]

Azure App Service Plan

Collection of integrated cloud services

Google Cloud Platform

An app service plan

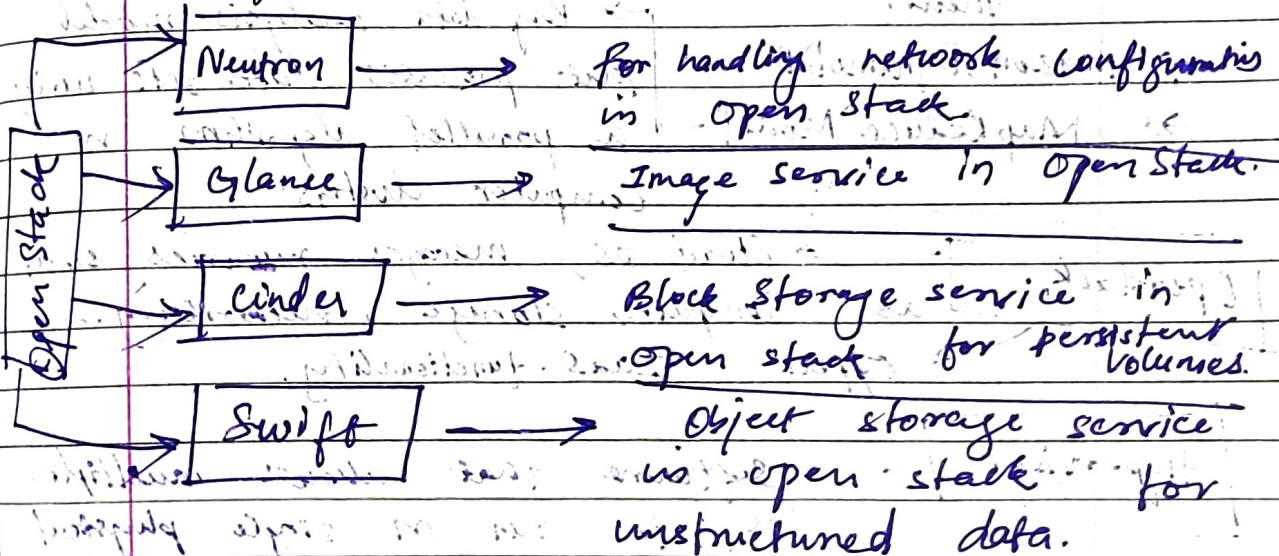
- * Region
- * Instance size
- * Scale count
- * SKU

Cloud service enables developers to build, test and deploy.

Compute
Storage
Services

1. Host your webpage in Google Cloud Platform
2. Build your web-app in Google App Engine.
3. Host your web-page in Google cloud platform

* Azure App has a scale count of 1 to 20 instances



Download Unit Cost = 120
→ for download of 100 units

Cinder manages block storage in OpenStack

Q. Peak computing demand = 100 units.

demand $\rightarrow D(t) = 50(1 + e^{-t})$

Baseline / owned

$\rightarrow$ unit cost = 120.

$\rightarrow$ cloud cost = 200.

Is cloud cheaper and which is cheaper for a period of 100 time units.

Sol

* $\rightarrow$ Owned unit cost = 120.

$\rightarrow$ for demand of 100 units,

total owned cost = 120×100

$\boxed{12000}$
units.

* Cloud cost:

= Actual usage $\times$ unit cloud cost

= $200 \int_0^{100} D(t) dt = 200 \int_0^{100} 50(1 + e^{-t}) dt$

= $10000 \int_0^{100} (1 + e^{-t}) dt = 10000 [t - e^{-t}]_0^{100}$

= $\boxed{1,010,000}$

$\rightarrow$ Owned = 12000 $\checkmark$: Cheaper.

Cloud = 1,010,000

Energy Conservation :

Resource management should be highly efficient. Reduce energy consumption.

Performance metrics for resource management:

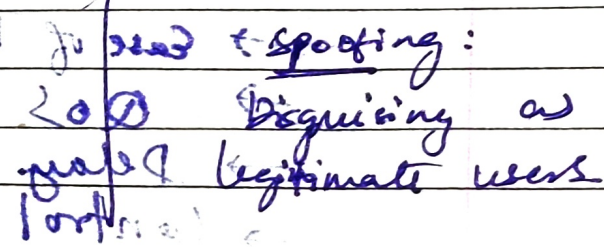
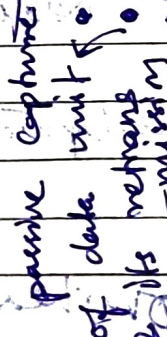
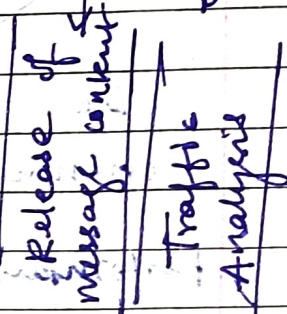
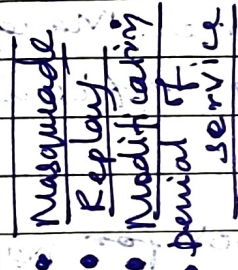
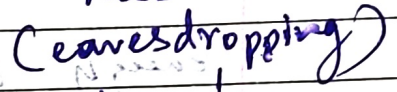
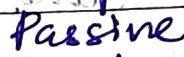
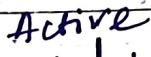
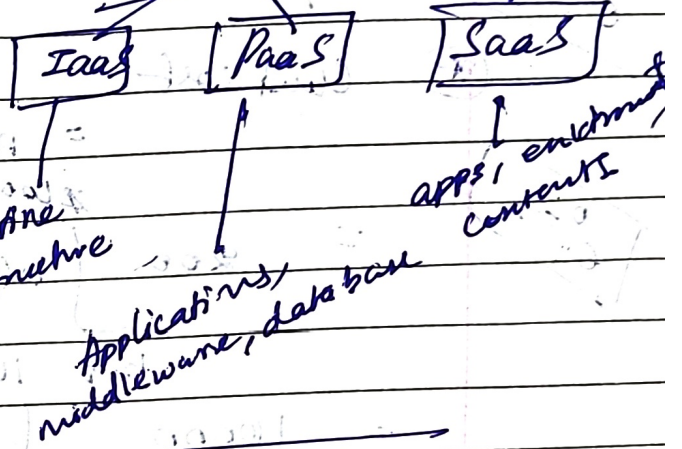
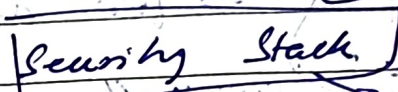
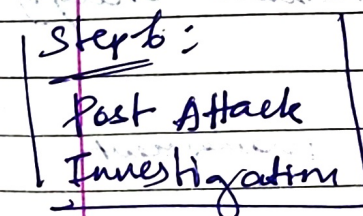
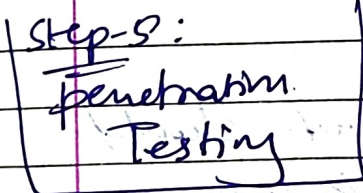
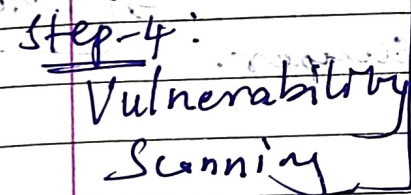
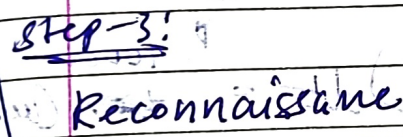
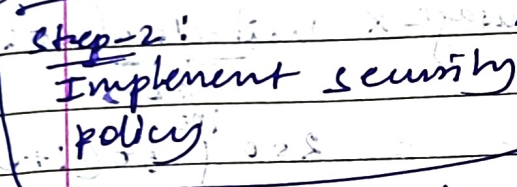
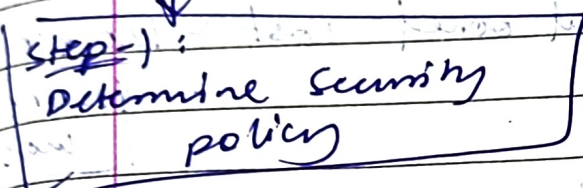
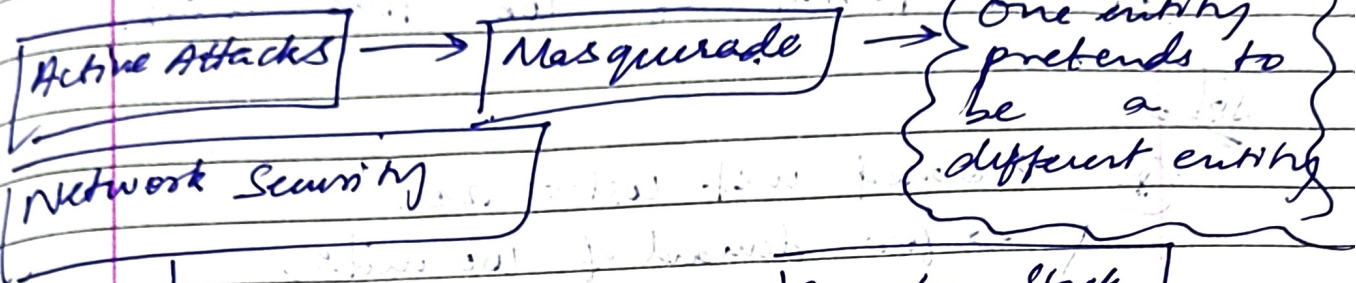
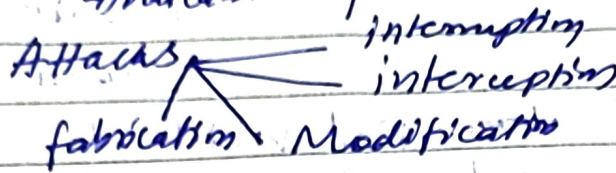
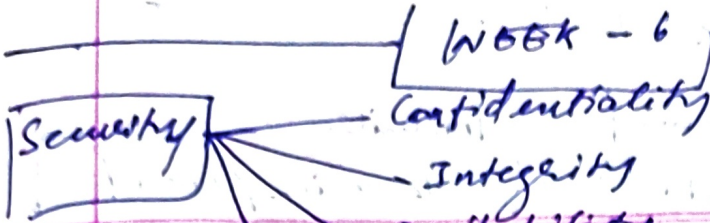
$\rightarrow$ Reliability

$\rightarrow$ Ease of deployment

$\rightarrow$ QoS

$\rightarrow$ Delay

$\rightarrow$ Control overhead.



Mobile Cloud Computing

Bringing rich computational resources to mobile users

Speed & flexibility

Shared resources

Integrated data

measures App speed, battery usage...

Dynamic Runtime Offloading

Profiling

Partitioning

separates apps that is to be run on cloud/mobile

migration

moves apps to cloud

Synchronization

in sync

Computing Offloading Approaches

→ static

→ Dynamic

→ Graph

Split apps before running

At runtime

Finding using best path using algorithms

→ * Energy - saving.

→ * Execution Time

Code offloading

Cloudlet

A small local server nearby a mobile device.

Used for computations.

* Mobile sends data to cloudlet for computation & also retrieves the results back. By this, power & time is conserved in a great way.

MUSIC framework

Mobility-aware Service Allocation

→ Monitor user location + movement

→ Smartly allocates cloud services based on location, power, delay cost.

Architecture

Amazon (AWS)

Azure

(Microsoft)

Tier-I

Local cloud

Tier-II

(nearby fast access)

Fog Computing: by Cisco ~~frameworks~~ systems

Computations brought to the closest or very nearby.

Reduces Latency. * IOT's

Example: * Smart Traffic Lights

* Connected Cars like Tesla.

Doesn't need to rely on cloud, as it is a "Fog system", can make decisions faster without relying on cloud.

Fog = handles local jobs & faster jobs

Cloud = Big and heavy jobs.

Geographic Information System [GIS]

→ They are tools that allow for the processing of spatial data into information.

→ allow to share services and data across organisations.

→ uses my cloud. No infrastructure required.

Geospatial Cloud:

It is a cloud computing platform that stores Geospatial data like maps, images, sensor data..., that are often huge in size.

Interoperability Issue:

→ Data-level
→ Service-level
→ Security level

Power Usage = $\frac{\text{Overall Power}}{\text{Power Delivered}}$

Power downtime = Migration time

migration time = $\frac{\text{VM size}}{\text{Transmission rate}}$

Docker

Development of Applications & shipping

Container Management Service

Abstraction @ App layer. It packages code & dependencies together

Application level Virtualization

Green Cloud

Impact of Cloud DC on Environment

A single host can run several spatial apps. for utilization of Resources

DC & Data centre

Sensor Cloud Computing

Virtual sensor
Weather monitoring system

eg: Traffic flow, temperature sensors, ...

Combines physical sensors such as temperature, Humidity, wind, ...

Image - Blueprint to create a Docker container.

consists of executable package that has all necessary items to run a piece of software

Built from a Dockerfile

contains instructions as text file

App-level Virtualization:

A single host can run several spatial apps for utilization of resources.

Virtual Sensors:

Weather system Monitoring

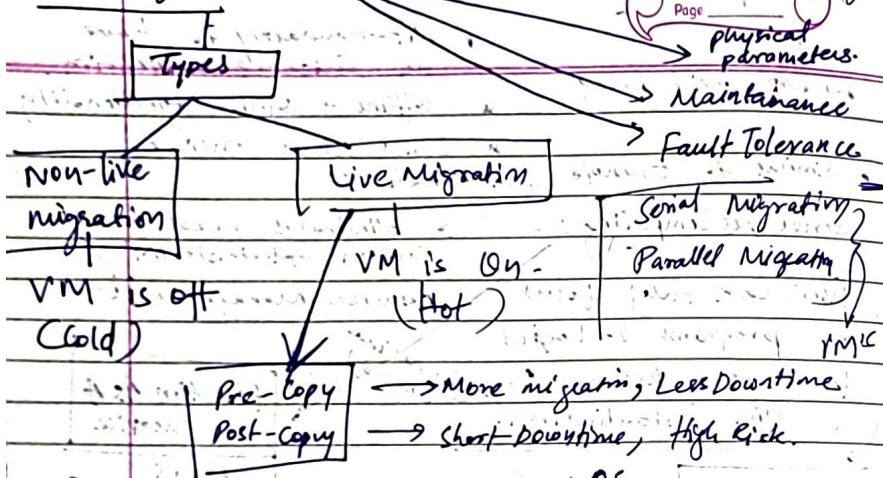
Sensor Cloud Computing

Green Cloud Computing

Power Usage effectiveness

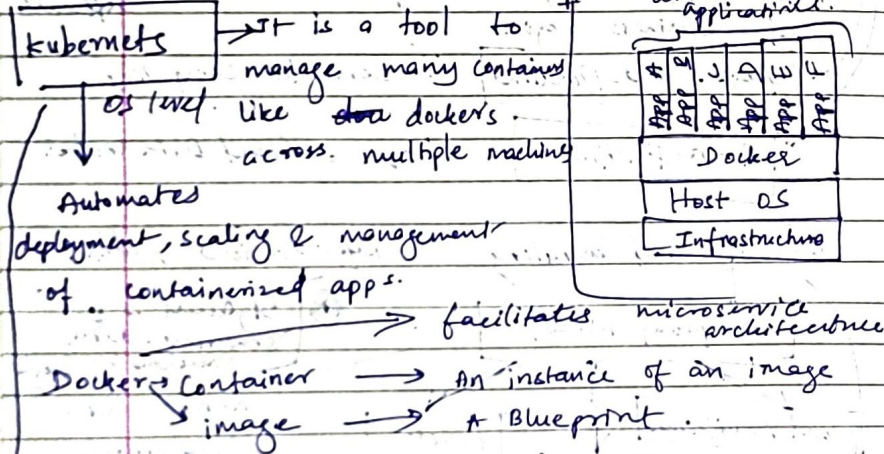
Overall Power Power Delivered

VM Migration:



Containers Virtualize @ the ~~Hardware~~ OS level
 VMs Virtualize @ the Hardware level

Kubernetes



Sometimes Docker are used to replace VMs like where the full focus is on application rather than the OS.

Kubernetes Cluster Components

- A Cluster - Control plane + Worker Nodes.
- Control Plane - Brain of the system, manages cluster
- Nodes - Run the app containers in pods.
- More Nodes - Better reliability & Handling

Worker nodes

Fog Computing

Delay Jitter

Variation in latency is observed while transmitting data over a network

→ Edge = Computing happens close to devices near where data is created.

→ Edge computing = Autonomous Vehicles (Sensor itself) Electric Vehicles

→ Fog Computing = Smart Traffic Management Systems. ≡ Smart Computers

connected with multiple sensors
 Computing

Edge (Decentralized) or Distributed Fog (Centralized)

Cloud-fog...
 Edge computing systems

Cloud Federation

Deployment & management of multiple external & internal cloud computing services to match business needs.

Federation
 ≡ A unit consists of small groups united to perform a common action.

Non-live Migration

$$\text{Total Migration Time} = \frac{\text{VM Size}}{\text{Transmission Rate}}$$

$$\text{Downtime} = \text{Total Migration Time}$$

Live Migration:

$$\text{Total Migration Time} = \frac{\text{VM Size} \times \text{no. of Iterations}}{\text{Transmission rate}}$$

$$\text{Downtime} = \frac{\text{Dirty pages} \times \text{Page Size}}{\text{T. Rate}}$$

$$\text{Expansion Time} = \frac{\text{Data Size}}{\text{Bandwidth}}$$

Edge Computing = Cloud Computing + End devices (Phone, PC, etc.)

Micro-service based

Not-mandatory to have an internet connection.

Offline + Online functionality

Syns when online

Similar to On-premise functionality

Serverless Computing:

Eliminates need of System Administration / Resource management

User can only focus on programs & Logics from the Resources or VM instantiation.

(Operational Logics)

Function As A Service

FaaS

eg: Payment, Clinical analysis

AWS Lambda

Serverless Computing Platform by AWS

Helps to focus only on core product like Avogadro Software and Business Logic

Google-Cloud-Functions

= Serverless Computing

Azure Functions

CDC = Cloud Data Centres

Sustainable Cloud Computing

Refer Previous Nodes

Conceptual Model

Cloud Architecture

Power Manager

Cooling Manager

BigQuery is a serverless-data warehouse by Google

Remote CDC

(Integrations)

(of VM's)

Reduce Carbon footprint (Green-house gases)

Cost Reduction

Emissions Reduction

Reduced Power Utilization

3 + objective

Amazon S3 Bucket

Simple Storage Service

Can store data in various formats

Cloud Computing in 5G

Approx. ~~date~~ ^{Deployment year} = 2020

Theoretical Down ~~Load~~ ^{Speed} = 10 GB/s.

Latency < 1ms

Mission critical Comms

Massive IoT Systems

Stringent QoS Requirements
A situation where cloud services ~~are~~ must meet high performance requirements for applications such as real-time comms or video-streaming.

Cyber-Physical System

Intersection of Cyber and Physical NAT Union

VM's instantaneous - AWS Lambda

Azure Function

Cyber-physical NAT Union

→ Fog computing
→ Edge computing

Integration of computers with physical processes; uses sensors, actuators to link computational system to physical world.

Spatial Cloud Computing:

* Maps, Visualisation, Locations & its data or information.

Google Cloud Functions

③ Trajectory cloud - Cloud-Based system that helps cities to understand how people move around.

Useful for Emergency services, Transport and delivery systems.

Integrated IoT

TS-1

TS-2

TS-3

Trajectory Service

→ India

→ US

→ Canada

GPS data

Map matching

Lets users ask questions

Privacy & Cloud sec

Takes the GPS route and the actual route

Google Cloud Engine

Google Compute Engine + Cloud SQL

Kubernetes

Amazon EC2 instances S3 Bucket

Full Revision

Logged Areas

manges VMs in Virtualization Environment

1. Hypervisor & its types.

2. Data Redundancy.

3. What & Why Amazon EC2
+ Azure + IBM cloud
+ Google App Engine

PaaS → Google App Engine
SaaS → Google Docs

4. API in Cloud Computing is used to enable comms b/w applications.

EC2 - By Amazon, for Managing VM's.
5. Docker - Containerization, Deployment/Managing Containers.

Xen - For Virtualization in Cloud Computing.
Kubernetes - Orchestration and management of containerized apps.

OpenStack - open source cloud Platform.

Cloud Monitoring tools - To monitor cloud services.

6. Multitenancy - multiple apps share same physical resources, but each is isolated.

7. AWS Lambda - Serverless service by AWS.

S3 - Bucket - Simple Storage Service.

8. Docker rmi - used to delete a local image.

9. Power Usage = Overall Power

Migration - Power Delivered.

Non-line: $\text{Total Mgr. time} = \frac{\text{VM size}}{\text{Transm. rate}}$

Line mgr: $\text{Total mgr. time} = \frac{\text{VM size} \times \text{no. of iterations}}{\text{Transm. rate}}$

Percentage - $\frac{\text{Dirty pages} \times \text{Page size}}{\text{Trans rate}}$

Azure - Hybrid Cloud Provider.

Cloud Bursting - Moving workloads to the cloud, during Peak usage

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AWS - offers BOTH IaaS and PaaS.

Elastic Load Balancing - Distributing incoming application traffic across multiple instances

Google BigQuery - Cloud-Based Big Data Processing service

Block-storage - data stored in ^{blocks} fixed size blocks for efficient storage & retrieval.

Cloud

Hypervisor type-1		Hypervisor type-2	
App	App	VM 1	VM 2
Guest OS	Guest OS	Hypervisor	Apps
Hypervisor		Host OS	
Hardware		Hardware	

Guest OS-1 Guest OS-2

Q. UDDI - Universal Description, Discovery & Integration
Directory service for discovering web services.

SLA - Service Level Agreement
Defines performance expectations.

SOAP - messaging protocol that allows different apps to communicate b/w

HTTP - Hyper text transfer protocol

Transfers data between web client & server

XML - extensible Markup language

Formats & transports structured data

Formulas

$$\text{Availability (A)} = 1 - \left(\frac{\text{downtime}}{\text{uptime}} \right)$$

$$1. \text{ Penalty} = 0.9 \cdot f \cdot \int \left| \frac{d}{dt} (b(t)) \right| dt$$

Longer
delay in
provisions

Change in
demand

$$2. \text{ Cloud Cost} = \text{Actual usage} \times \text{Unit Cloud Cost}$$

$$3. \text{ Power Usage} = \frac{\text{Overall Power}}{\text{Power Delivered}}$$

$$\text{Migration Time} = \frac{\text{VM Size}}{\text{Transmission Rate}}$$

SLA contains SLO -